



SIMATS ENGINEERING
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Course Code: CSA 0504

Slot:D

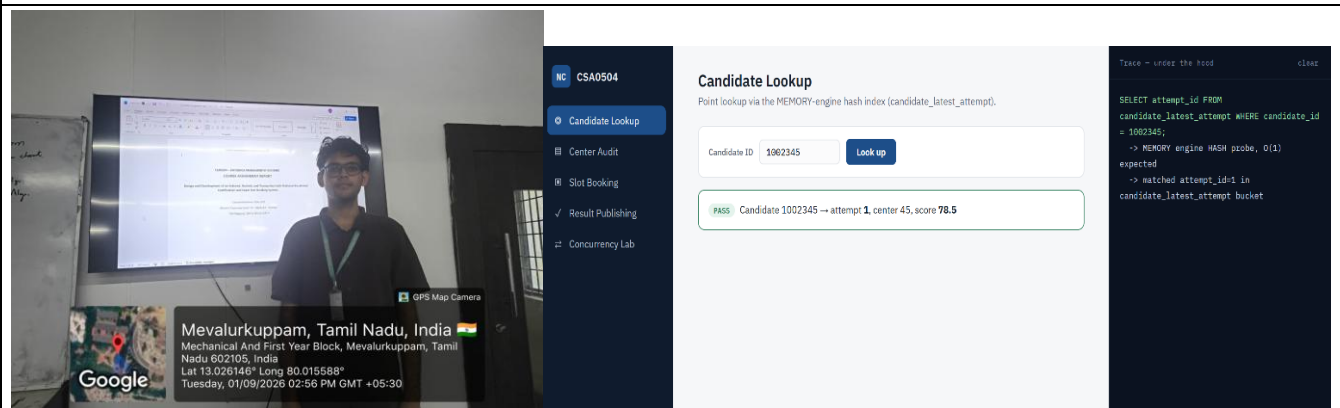
Course Name: DATABASE MANAGEMENT SYSTEMS

Course Faculty: Dr. R. Kesavan

Project Title:

National Vocational Certification and Exam Slot-Booking System: File Organization, Indexing, Hashing, Query Optimization, and Transaction Management

Module Photographs: (3 photographs –Module Photo, Individual student contribution module work in the project and presentation image)



Project Description: (here you write what you did in this project (contribution) including Model Description)

This project addressed two reliability problems in a national vocational certification exam system: slow retrieval from a single, ever-growing sequential archive of exam-attempt records, and unsafe concurrent access during exam-slot booking and result publishing. For the archive, an indexed-sequential file organization was implemented in MySQL (InnoDB) using a clustered primary key together with B-tree secondary indexes on candidate_id and on (center_id, exam_date), supplemented by a MEMORY-engine hash table for O(1) candidate lookup, and year-based range partitioning for further pruning. For booking and results, a relational schema (exam_slots, bookings, results) was designed and made transaction-safe: slot booking uses a bounded transaction with SELECT ... FOR UPDATE and a SAVEPOINT to guarantee a slot is never booked beyond capacity, while result publishing uses optimistic concurrency control via a version column so that simultaneous updates by different examiners are detected and retried instead of silently overwriting one another. Query execution plans were analysed with EXPLAIN before and after adding a covering index on the results table to remove a full-table-scan join. Concurrency was validated by running two simulated sessions against the same slot and the same result row and confirming that capacity was never exceeded and no update was lost.

Student Signature

Guide Signature